MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS) FIRST SEMESTER EXAMINATION

(2024ADMISIONONWARDS)

MG2DSCZGY101 – Biological Basis of Behaviour II

Practical Model Question Paper

Duration:2hrs MaximumMarks:35

Certified Record 10 mark

1. Identify the labelled parts and comment on the molecular composition of DNA. 5 Marks

a) Identify any 4 parts – 2 marks

b) comment -3 marks (CO: 5) (Understand)

2. Identify and comment on any two stages of mitosis - 2 marks

a) Identification: 1 mark
b) Comment: 1 mark (CO: 5) (Understand)

3. Identify and comment on any two - mendelian disorders/karyotype of Chromosomal disorders/normal karyotype of human - 4 Marks

a) Identification: 2 marks
b) Comment: 2 marks (CO: 5) (Understand)

4. Solve the given genetic problem (monohybrid cross/ dihybrid cross/ test cross/ back cross) - 8 Marks (CO: 5) (Apply)

Example: In pea plants, round seed shape (R) is dominant over wrinkled (r), and yellow seed colour (Y) is dominant over green (y). A plant that is heterozygous for both traits (RrYy) is crossed with another plant that is also heterozygous for both traits (RrYy).

(a) Construct a Punnett square for this dihybrid cross.
(b) Determine the phenotypic ratio of the offspring.
(c) How many offspring are expected to have round yellow, round green, wrinkled yellow, and wrinkled green seeds?

5. Identify and comment on symbols in pedigree chart (Any two) -2 Marks

a) Identification: 1 mark
b) Comment: 1 mark (CO: 5) (Understand)

6. Construct a pedigree chart for the given inheritance (Haemophilia) - 4 Marks (CO: 5) (Analyse)

Example: A man with haemophilia marries a woman who is a carrier for the condition. They have three children: a healthy son, a daughter who is a carrier, and a son with haemophilia. Construct a pedigree chart representing this family's inheritance pattern.

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